



ANDISHE SHOMAL

Andisheh Shomal Machine Manufacturing Company

Internship Project Report

Title

Finite Element Analysis and Stress Investigation of the Mesh Filter in the Scrubber Device

Intern:

Mohammadreza Sotoodeh

Internship Supervisors:

Ms. Engineer Hosseinpour

Ms. Engineer Nasiri

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Abstract

In this project, the mechanical behavior and the stress levels generated in the mesh filter used in the scrubber device under compressive loading were investigated. Due to the mesh geometry of the filter and the presence of stress concentration zones, simple analytical methods were insufficient for accurately evaluating the structural performance. Therefore, Finite Element Analysis (FEA) was employed to examine the stress distribution and assess the strength of the component. The geometric model of the filter was created in the SolidWorks environment, and numerical analysis was performed using SolidWorks Simulation. A working pressure of 7 bar was applied as the primary loading on the filter surface, and the von Mises stress distribution within the structure was examined. These results enabled the identification of critical regions and a preliminary evaluation of the mechanical performance of the filter under operating conditions. Ultimately, finite element analysis was utilized as an effective tool for assessing strength, identifying critical points, and assisting in the optimization of the mesh filter design.

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1 Introduction

Mesh filters are among the important components in certain industrial equipment, including scrubber devices, and are responsible for separating, allowing passage, or distributing fluid flow and particles. During operation, these components may be subjected to internal pressure or pressure differences across the filter. Therefore, their proper design from the standpoint of mechanical strength is of great importance, since the presence of holes, abrupt geometric changes, and limited thickness can lead to stress concentrations and increase the likelihood of deformation or damage under working conditions.

In this project, Finite Element Analysis (FEA) was performed to investigate the mechanical performance of the mesh filter of the scrubber device. The finite element method is one of the widely used numerical methods in mechanical engineering that enables the examination of the behavior of structures and components with complex geometries under various loading conditions. In this method, the geometry of the component is divided into a large number of small elements, and by numerically solving the governing equations, quantities such as displacement, strain, and stress are calculated at different points of the component.

In this study, the geometric model of the filter was prepared in the SolidWorks environment and then, using the SolidWorks Simulation module, the desired loading conditions and constraints were defined. A pressure of 7 bar was considered as the applied pressure on the filter, and the von Mises stress distribution in different parts of the component was evaluated. The main focus of the analysis was on identifying the regions with the highest stress and comparing the behavior of two filter samples under identical conditions.

The objective of performing this analysis, in addition to examining the stress levels generated in the components, was to identify critical points and provide a basis for evaluating strength and improving the filter design. The results obtained can be used in decision-making related to design, dimensional selection, and assessment of the component's performance under operating conditions.

2 Finite Element Analysis and Stress Investigation of the Mesh Filter in the Scrubber Device

One of the subjects investigated during the internship period was the mesh filter used in the gas scrubber device. This filter consists of a perforated metal sheet that, in addition to providing a flow path, must also possess sufficient mechanical strength against the operating pressure of the equipment.

The geometry and arrangement of the holes created in the sheet can have a significant effect on its mechanical behavior. Creating holes reduces the effective cross-sectional area of the material and also introduces geometric discontinuities that can cause stress concentration around the holes. Therefore, the appropriate selection of hole diameter and the spacing between holes is also important from a mechanical perspective.

In order to investigate the effect of the perforation geometry on the mechanical behavior of the filter, two different designs were modeled in SolidWorks software and then subjected to Finite Element Analysis (FEA) using the SolidWorks Simulation module.

The main objective of this analysis was to determine and compare the maximum von Mises equivalent stress in the two designs under the same operating pressure.

2.1 Specifications of the Filter under Study

The filter under study is designed as a cylindrical shell made of a metal sheet. The overall geometric and technical specifications of the model under investigation are as follows:

Parameter	Value
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Overall shape	Cylindrical
Filter diameter	100 mm
Filter height	300 mm
Sheet thickness	0.8 mm
Sheet material	Galvanized Carbon Steel
Working pressure	7 bar

In the filter design, the cylindrical surface is covered with a specific pattern of holes. In order to investigate the effect of hole size and spacing on mechanical behavior, two different arrangements were studied.

2.2 Objectives of the Analysis

The finite element analysis was performed with the following objectives:

1. Investigation of the mechanical behavior of the filter under working pressure.
2. Determination of the maximum von Mises equivalent stress in each of the designs.
3. Examination of the effect of hole size and Pitch on stress distribution.
4. Identification of regions with the highest stress concentration.
5. Comparison of the mechanical performance of the two different designs.
6. Providing a basis for the selection and optimization of a suitable filter geometry.

Since the loading conditions, material, and sheet thickness were considered identical in both models, comparison of the results can provide appropriate information about the effect of hole geometry on the mechanical behavior of the filter.

2.3 Definition of the Cases under Investigation

For the purpose of comparison, two models with different perforation patterns were considered. In both models, the overall dimensions of the filter, the material and thickness of the sheet, and the loading conditions were identical.

The main difference between the two models was in the **hole diameter and the center-to-center spacing of the holes (Pitch)**.

Case 1

In the first model, holes with a diameter of **3 mm** and a Pitch of **5 mm** were considered.

Specifications of Case 1:

- Hole diameter: 3 mm
- Pitch: 5 mm
- Sheet thickness: 0.8 mm
- Material: Galvanized Carbon Steel
- Filter diameter: 100 mm
- Filter height: 300 mm
- Working pressure: 7 bar

In this arrangement, the holes have smaller dimensions and their center-to-center spacing is also smaller.

Case 2

In the second model, the hole diameter was increased to **5 mm** and the Pitch was set to **8 mm**.

Specifications of Case 2:

- Hole diameter: 5 mm
- Pitch: 8 mm
- Sheet thickness: 0.8 mm

- Material: Galvanized Carbon Steel
- Filter diameter: 100 mm
- Filter height: 300 mm
- Working pressure: 7 bar

In this case, the holes are larger than in Case 1 and their center-to-center spacing is also increased.

2.4 Geometric Modeling

The three-dimensional model of the filter was created in the SolidWorks environment. First, the overall cylindrical geometry with a diameter of 100 mm, a height of 300 mm, and a thickness of 0.8 mm was generated. Then, the desired perforation pattern was created on the sheet surface. For Case 1, 3 mm holes with a Pitch of 5 mm were used, and for Case 2, 5 mm holes with a Pitch of 8 mm were employed.

After completing the three-dimensional models, both models were imported into the SolidWorks Simulation environment for mechanical analysis.

2.5 Material Properties Definition

For both models, the sheet material was considered to be Galvanized Carbon Steel. Using the same material in both analyses ensured that the comparison of results was not influenced by changes in material properties and that the differences in results were primarily related to differences in geometry and hole arrangement.

The mechanical properties required by the software for performing the analysis were used based on the material defined in the model.

2.6 Loading Conditions Definition

Since the objective of the analysis was to examine the performance of the filter under the operating conditions of the equipment, a working pressure of 7 bar was considered as the analysis loading.

This pressure was applied identically for both Case 1 and Case 2 models. Therefore, the loading conditions of the two analyses were completely similar:

Pressure = 7 bar

Applying the same pressure to both models made it possible to directly compare the effect of differences in the perforation geometry on the mechanical behavior of the filter.

2.7 Boundary Conditions

In addition to applying the working pressure, the boundary conditions of the model were also defined in accordance with the manner in which the filter is positioned in the equipment. Parts of the model that are connected to or constrained by other components of the equipment under real conditions were constrained in the analysis in such a way as to prevent rigid-body motion of the model and to simulate the working conditions of the filter as closely as possible.

Correct definition of boundary conditions is one of the important factors in finite element analysis, because the application of inappropriate constraints can lead to unrealistic results in stress distribution and deformation.

2.8 Model Meshing

After defining the geometry, material, boundary conditions, and working pressure, the model was meshed in the SolidWorks Simulation environment. Given the large number of holes and the geometric variations around them, the regions surrounding the holes were of particular importance.

These regions are the locations of force transmission path changes and the generation of stress concentrations.

For this reason, mesh quality around the holes is of great importance for the accuracy of the analysis results. After creating the mesh, the analysis was performed separately for each of the two models.

2.9 Stress Evaluation Criterion

To examine the stress state in the filter, the Von Mises Equivalent Stress criterion was used. The von Mises equivalent stress is one of the common criteria for evaluating the likelihood of yielding in metallic materials under multiaxial loading conditions. The value of this stress was calculated at different points of the model, and its maximum value was extracted as the Maximum Von Mises Stress.

Given that the two models were analyzed under identical loading conditions, comparison of the maximum von Mises stress can serve as a suitable criterion for examining the effect of hole geometry on the mechanical behavior of the filter.

2.10 Comparison of Analysis Results

The final results of the two analyses are presented in the table below:

Parameter	Case 1	Case 2
Hole diameter	3 mm	5 mm
Pitch	5 mm	8 mm
Thickness	0.8 mm	0.8 mm
Working pressure	7 bar	7 bar
Maximum Von Mises Stress	5.2×10^7 Pa	7.2×10^7 Pa
Maximum stress	52 MPa	72 MPa

Based on the results obtained, the maximum stress value in Case 2 is higher than in Case 1.

The percentage increase in maximum stress is calculated from the following relation:

$$\text{Percentage Increase} = [(72 - 52) / 52] \times 100$$

Therefore:

$$\text{Percentage Increase} \approx 38.5\%$$

As a result, the maximum von Mises equivalent stress in Case 2 is approximately **38.5 percent higher than in Case 1**.

2.11 Interpretation of Results

The analysis results indicate that changes in the perforation geometry can have a significant effect on the stress distribution in the filter.

In both models, the working pressure, material, thickness, and overall dimensions of the filter were identical; however, changes in hole diameter and Pitch resulted in differences in the maximum stress value.

In Case 1, the maximum stress was 52 MPa, and in Case 2 it was 72 MPa. The increase in stress in Case 2 can be associated with the change in the local geometry of the sheet around the holes and the alteration of the force transmission path.

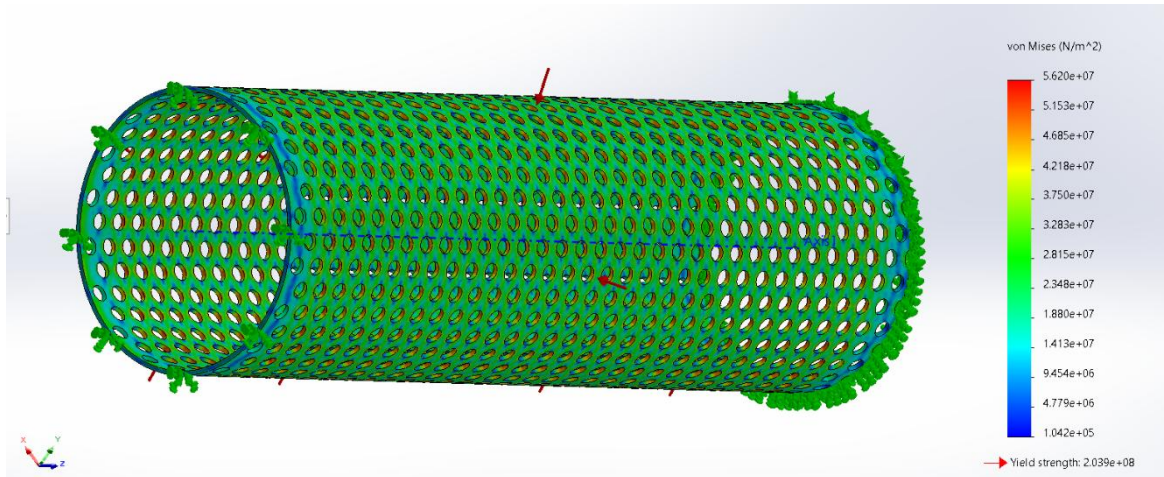
Holes create discontinuities in the sheet, and stress concentration forms around their edges. Changes in hole diameter and the spacing between them alter the amount of remaining material between the holes and the manner of force distribution in these regions.

Therefore, the selection of the perforation pattern should not be based solely on parameters such as open area or flow-passing capability; the mechanical behavior of the filter must also be taken into account in the design process.

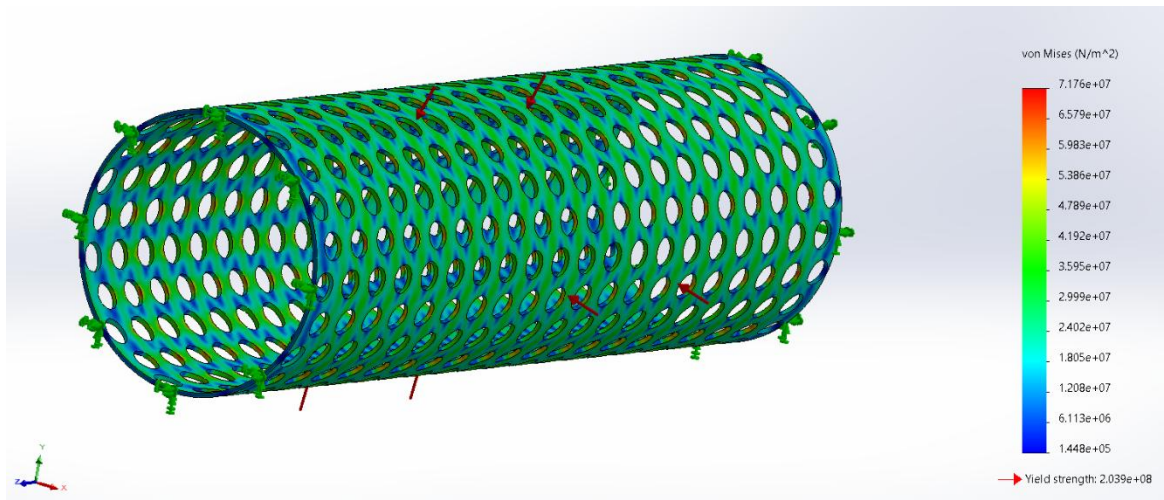
2.12 Presentation of Stress Analysis Results

In this section, images obtained from the finite element analysis in SolidWorks Simulation software are presented. The color contours indicate the distribution of the von Mises equivalent stress (Von Mises Stress) on the filter surface under a working pressure of 7 bar.

Von Mises Stress Distribution in Case 1



Von Mises Stress Distribution in Case 2



3 Conclusion

In this project, in order to investigate the mechanical strength of the mesh filter of the scrubber device, finite element analysis was performed using SolidWorks Simulation software. The filter was examined under a pressure of 7 bar, and the von Mises stress distribution at various points was obtained. Given the mesh geometry of the component, the regions around the holes and geometric changes were considered as points susceptible to stress concentration.

The analysis results showed that the maximum von Mises stress values in the two samples under investigation were approximately 52 MPa and 72 MPa, respectively. Therefore, the first sample experienced a lower maximum stress compared to the second sample and, from the viewpoint of stress distribution, exhibited more favorable performance under the loading conditions examined. Nevertheless, for a definitive judgment regarding the safety and usability of the component,

comparison of the obtained stresses with the yield strength and the factor of safety of the material used is essential.

In general, performing finite element analysis made it possible to identify the critical points of the filter and to examine its behavior under working pressure. The results of this study can be used as a basis for modifying the geometry, reducing stress concentrations, and optimizing the filter design. Furthermore, this project demonstrated that the use of numerical analysis prior to manufacturing or modifying components can be effective in reducing trial-and-error costs and increasing confidence in the mechanical performance of industrial equipment.